

# HSE711\_Final\_Project

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```
metadata.modified <- meta_data.df %>%
  dplyr::select(1,10,23,29,41) %>%
  mutate(barcode = str_replace_all(barcode, "-", "."))

# Convert count.df to long format

dat.long <- count.df %>%
  pivot_longer(
    cols = -X,          # Pivot all columns except the new row_id column
    names_to = "Samples", # Column name for your old wide headers
    values_to = "Counts" # Column name for your data values
  )

dat.long.enriched <- dat.long %>%
  mutate(X = sub("\\.[^.]*$", "", X)) %>%
  inner_join(., metadata.modified, by = c("Samples" = "barcode"))

library(AnnotationDbi)
library(org.Hs.eg.db)

count.df.mod <- count.df %>%
  mutate(X = sub("\\.[^.]*$", "", X))

ensemble.ids <- count.df.mod %>% pull(X)

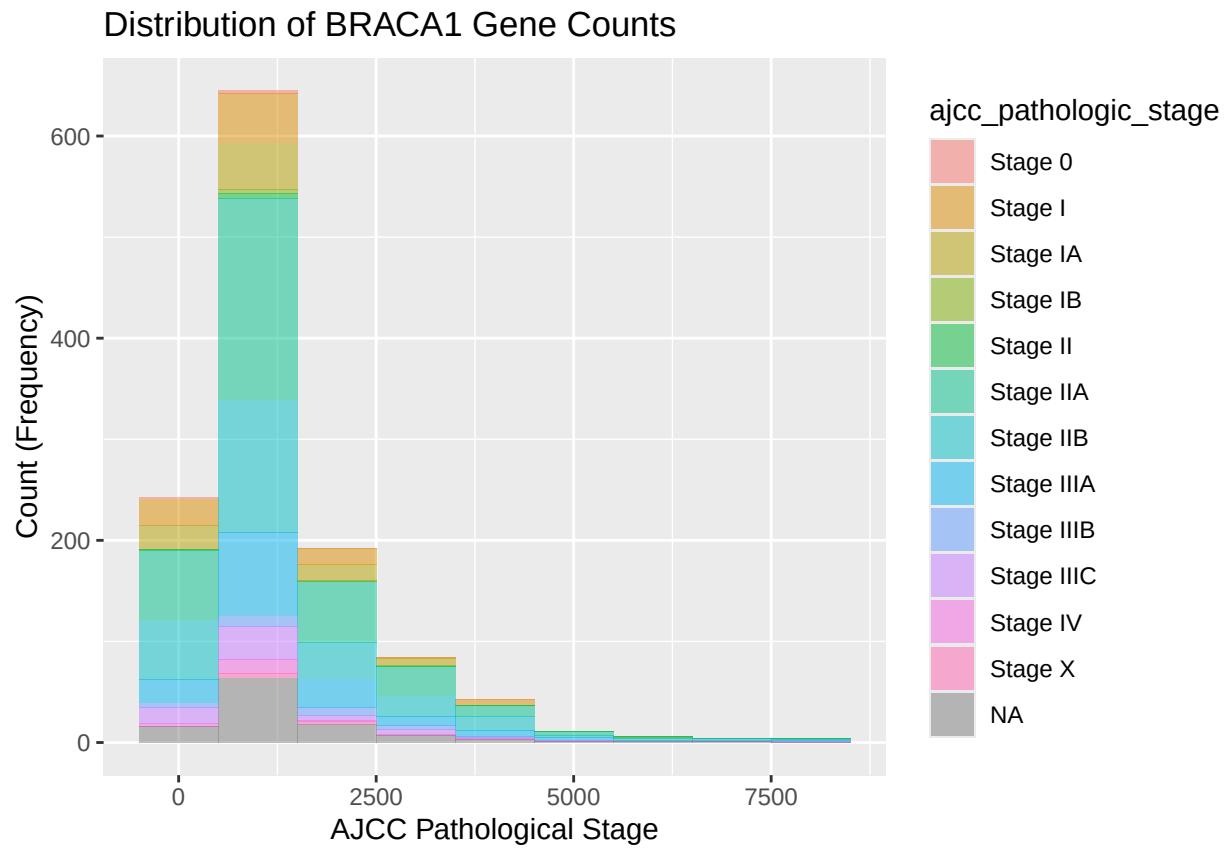
ensembl_ids <- ensemble.ids

# Map IDs to Gene Symbols
gene_symbols <- mapIds(
  x = org.Hs.eg.db,
  keys = ensembl_ids,
  keytype = "ENSEMBL",
  column = "SYMBOL",
  multiVals = "first" # Keeps the first match if duplicate mappings occur
)

df <- data.frame(names(gene_symbols), gene_symbols)

dat.long.genes <- dat.long.enriched %>%
  # mutate(X = sub("\\.[^.]*$", "", X)) %>%
  inner_join(., df, by = c("X" = "names.gene_symbols."))
```

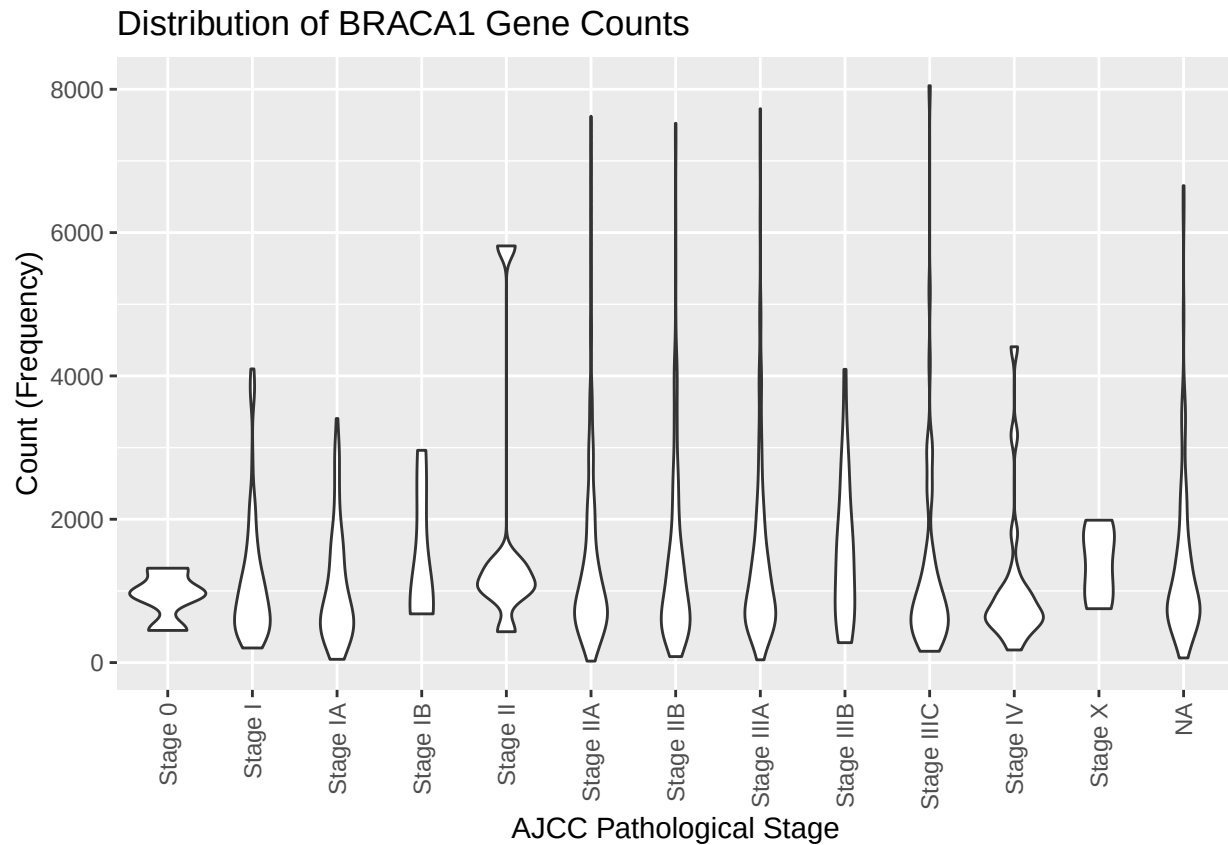
```
# 1. barplot
dat.long.genes %>%
  filter(gene_symbols == 'BRCA1') %>%
  ggplot(., aes(x=Counts, fill = ajcc_pathologic_stage)) +
  geom_histogram(alpha = 0.5, binwidth = 1000) +
  labs(
    title = "Distribution of BRCA1 Gene Counts",
    x = "AJCC Pathological Stage",
    y = "Count (Frequency)"
  )
)
```



```

dat.long.genes %>%
  filter(gene_symbols == 'BRCA1') %>%
  ggplot(., aes(x=ajcc_pathologic_stage,y=Counts)) +
  geom_violin() +
  labs(
    title = "Distribution of BRCA1 Gene Counts",
    x = "AJCC Pathological Stage",
    y = "Count (Frequency)"
  ) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))

```

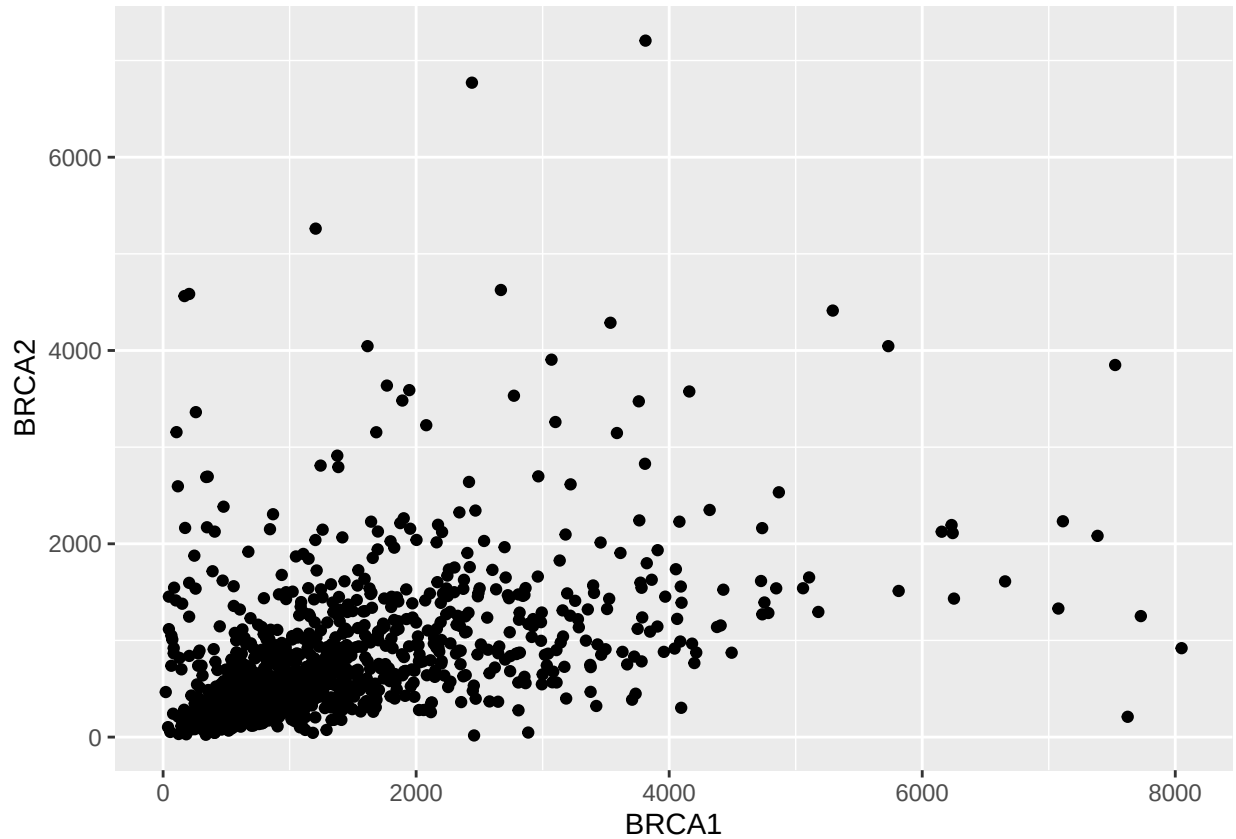


```
summary_stats <- dat.long.genes %>%
  filter(gene_symbols == 'BRCA2') %>%
  summarize(
    min = min(Counts, na.rm = TRUE),
    max = max(Counts, na.rm = TRUE),
    mean = mean(Counts, na.rm = TRUE),
    median = median(Counts, na.rm = TRUE),
    sd = sd(Counts, na.rm = TRUE)
  )
summary_stats
```

```
## # A tibble: 1 x 5
##   min    max  mean median    sd
##   <int> <int> <dbl>  <int> <dbl>
## 1     17  7206  749.    529  717.
```

```
genes.of.interest <- c('BRCA1', 'BRCA2')

#df_test <- dat.long.genes %>%
dat.long.genes %>%
# filter(gene_symbols == 'BRCA1' | gene_symbols == 'MYCBP2') %>%
  filter(gene_symbols %in% genes.of.interest) %>%
  dplyr::select(-1) %>%
  pivot_wider(names_from = gene_symbols, values_from = Counts) %>%
  ggplot(., aes(x=BRCA1,y=BRCA2)) +
  geom_point()
```



```
#df_test <- dat.long.genes %>%
# filter(gene_symbols %in% genes.of.interest) %>%
# dplyr::select(-1) %>%
# pivot_wider(names_from = gene_symbols, values_from = Counts)
```